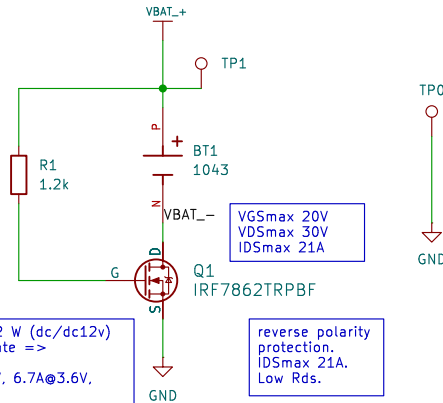
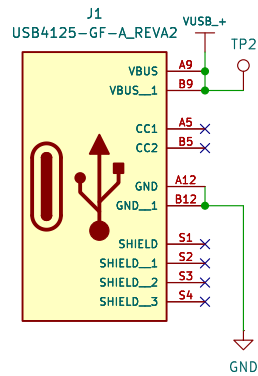
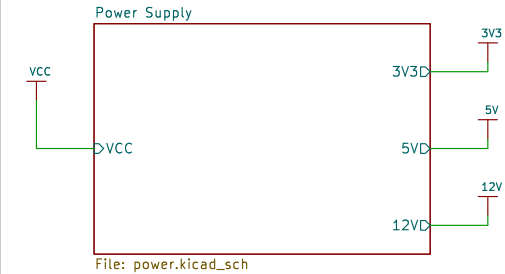


## Supply connector

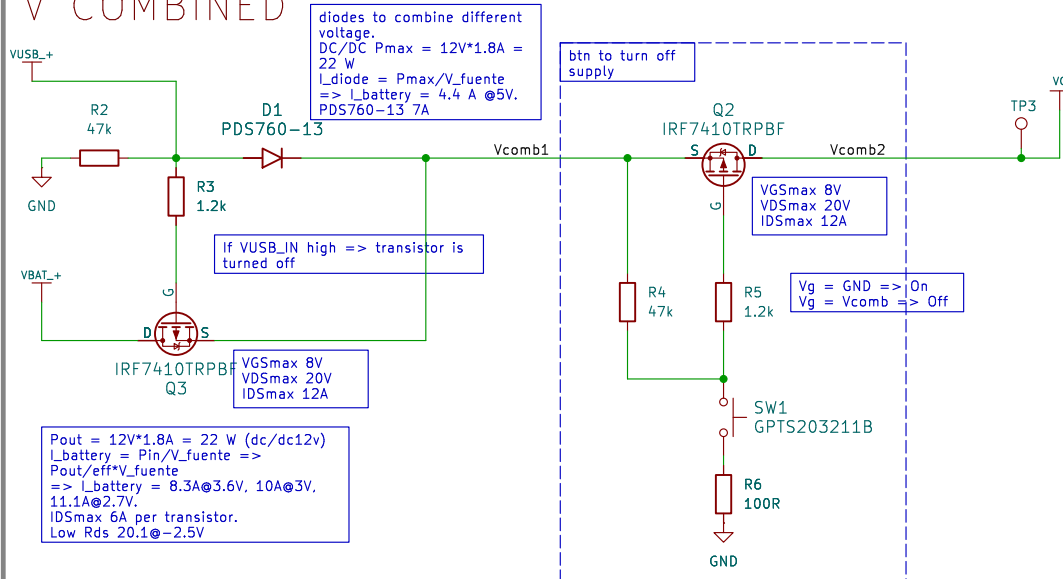


$P_{out} = 12V \cdot 1.8A = 22 \text{ W (dc/dc12v)}$   
 $I_{battery} = P_{in}/V_{fuente} \Rightarrow$   
 $P_{out}/\text{eff} \cdot V_{fuente}$   
 $\Rightarrow I_{battery} = 8A@3V, 6.7A@3.6V,$   
 $8.9A@2.7V.$

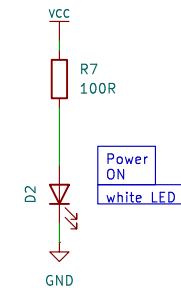
## Power Supply



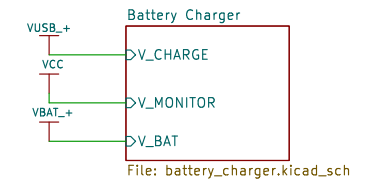
## V COMBINED



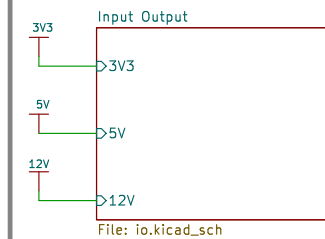
## LEDs



## Battery charger



## Input Output



Sheet: /  
 File: pcb\_microbit.kicad\_sch

**Title:**

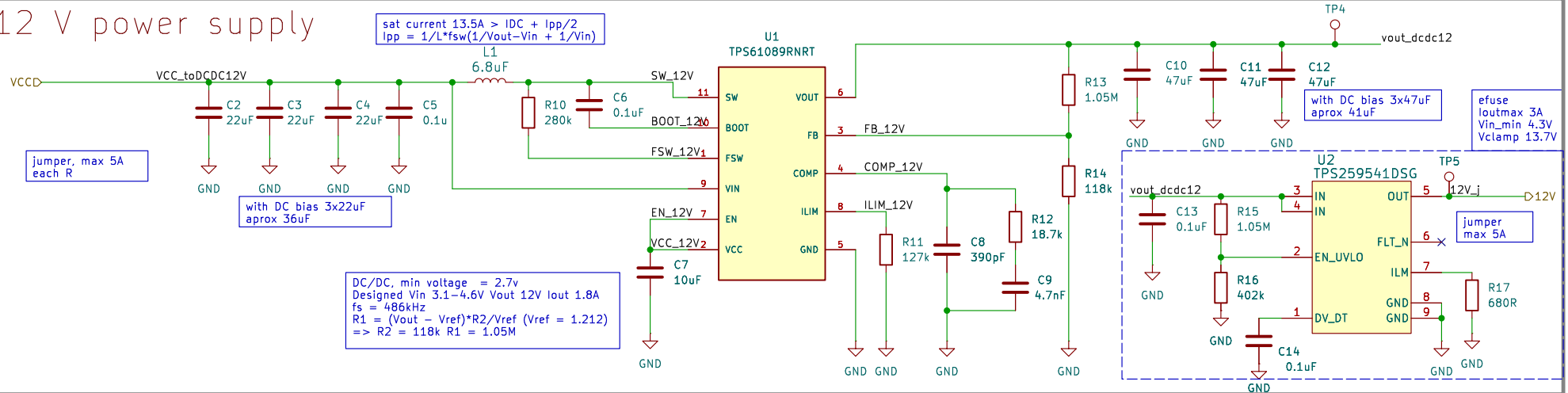
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Date:

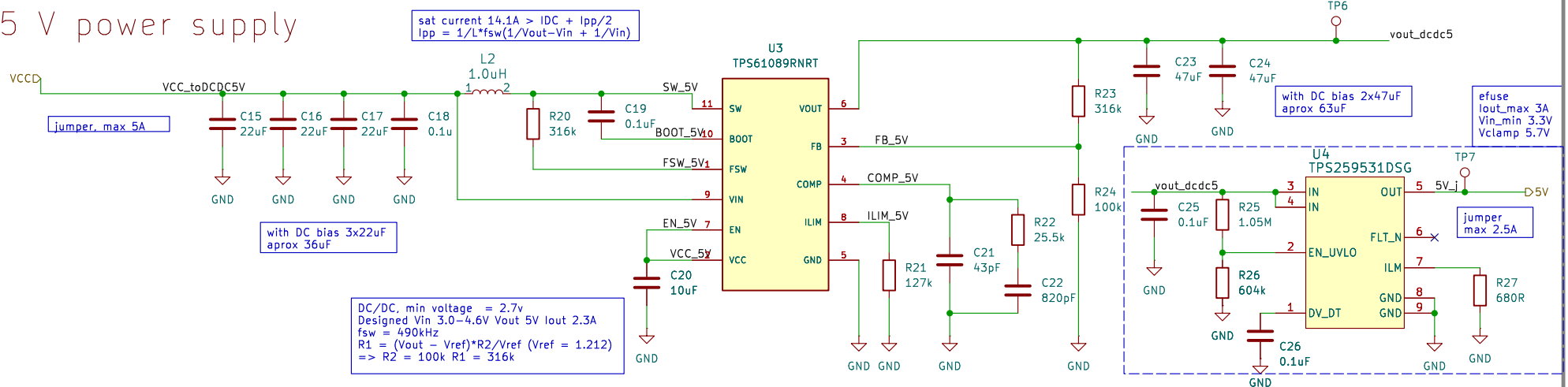
**Rev:**

Id: 1/4

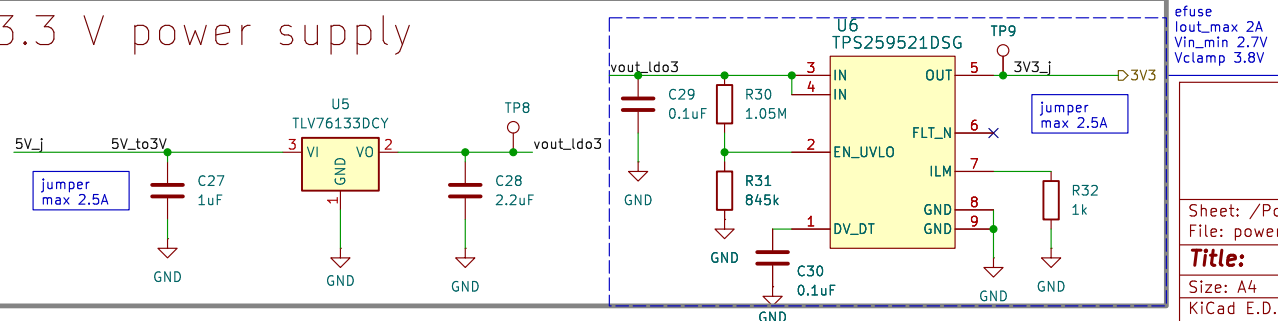
## 12 V power supply



## 5 V power supply



## 3.3 V power supply



Sheet: /Power Supply/  
 File: power.kicad\_sch

Title:

Size: A4

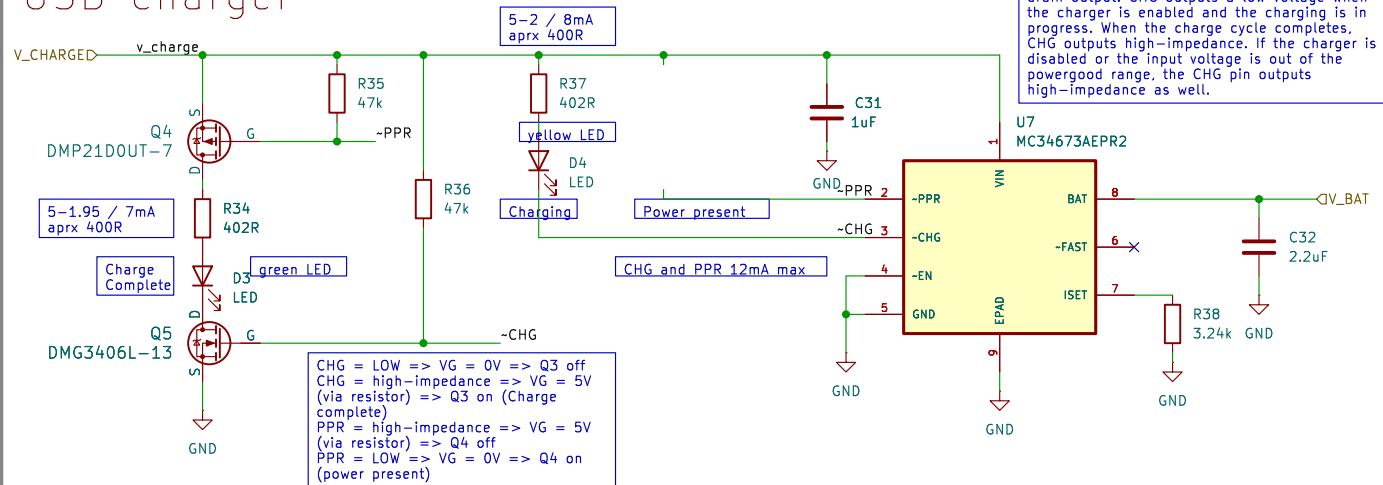
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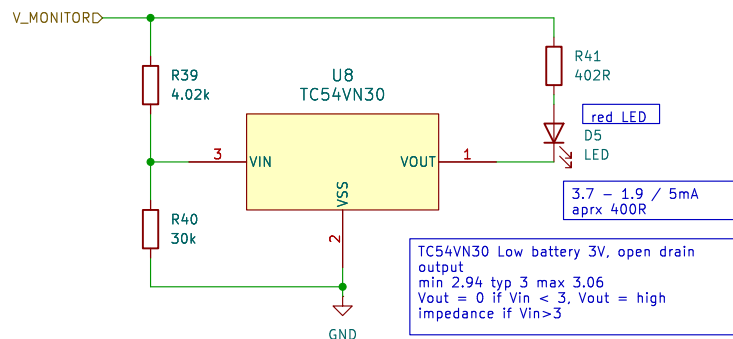
Rev:

Id: 2/4

## USB charger



## Voltage monitor



V\_BAT > V\_DET = 3.0 TC54VN30  
 $R1 + R2 = 3.4 / 100\mu A \Rightarrow R1 + R2 = 34k$   
 $3.0 / 3.4 = 0.88$   
 $R2 / (R1 + R2) = 0.88$   
 $R2 / 34k = 0.88 \Rightarrow R2 = 29.9k$   
 $R1 = 4.8k$   
 $R1 = 4k, R2 = 30k \Rightarrow V_{bat\_low} = 3.4 * R2 / R1 + R2 = 3.0V$

Sheet: /Battery Charger/  
 File: battery\_charger.kicad\_sch

### Title:

Size: A4

Date:

KiCad E.D.A. 8.0.9

Rev:

Id: 3/4

U10  
PCA9685PW.118

3V3\_in

C33  
0.1uF

GND

28 VDD

23 -OE

25 EXTCLK

1 A0

2 A1

3 A2

4 A3

5 A4

24 A5

14 VSS

GND

27 SDA

26 SCL

6 PWM3

7 PWM2

8 PWM1

9 PWM4

10 PWM5

11 PWM6

12 LED0

13 LED1

15 LED2

16 LED3

17 LED4

18 LED5

19 LED6

20 LED7

21 LED8

22 LED9

24 LED10

The diagram shows a motor driver circuit using a TB6612FNG IC. The IC is powered by a 3V3\_in and 12V\_in. It controls two motors, MOTOR1\_1 and MOTOR2\_1, using PWM signals from the Arduino. The circuit includes current sense resistors R42 and R43, and green LEDs for status indication. The motor current is measured to be 1.2-1.95 A at 7mA.

| Pin  | Label                        |
|------|------------------------------|
| P3   | 1 reserved LED matrix        |
| P2   | 2 reserved LED matrix        |
| P0   | 3 PIN_AD_0                   |
| P0   | 4                            |
| P0   | 5                            |
| P4   | 6 reserved LED matrix        |
| P5   | 7 reserved button B          |
| P6   | 8 reserved LED matrix        |
| P7   | 9 reserved LED matrix        |
| P1   | 10                           |
| P1   | 11 PIN_AD_1                  |
| P1   | 12                           |
| P1   | 13                           |
| P8   | 14 PIN_PWM_MOTOR2            |
| P9   | 15 PIN_D_9                   |
| P10  | 16 reserved LED matrix       |
| P11  | 17 reserved button A         |
| P12  | 18 PIN_D_12 reserved         |
| P2   | 19                           |
| P2   | 20 reserved LED matrix in v1 |
| P2   | 21                           |
| P2   | 22                           |
| P2   | 23 PIN_STBY_MOTOR            |
| P13  | 24 PIN_DIR_MOTOR1            |
| P14  | 25 PIN_PWM_MOTOR1            |
| P15  | 26 PIN_DIR_MOTOR2            |
| P16  | 27 3V3_microbit              |
| 3.3V | 28 3V3_microbit              |
| 3.3V | 29 3V3_microbit              |
| 3.3V | 30 3V3_microbit              |
| 3.3V | 31 3V3_microbit              |
| 3.3V | 32 3V3_microbit              |
| 3.3V | 33 SCL                       |
| P19  | 34 SDA                       |
| P20  | 35 GND                       |
| GND  | 36 GND                       |
| GND  | 37 GND                       |
| GND  | 38 GND                       |
| GND  | 39 GND                       |
| GND  | 40 GND                       |

[illegible]

The schematic shows the I2C connection between the ADS1015 (U14) and PCA9536 (U15) modules. The ADS1015 has pins for ADDR (1), AIN0 (4), AIN1 (5), AIN2 (6), AIN3 (7), SCL (10), and SDA (9). The PCA9536 has pins for VCC (8), SCL (6), SDA (7), and four output pins (P0, P1, P2, P3) labeled 1, 2, 3, and 5 respectively. The connections are as follows: ADDR (1) is connected to VDD (8); AIN0 (4) is connected to VDD (8); AIN1 (5) is connected to VDD (8); AIN2 (6) is connected to VDD (8); AIN3 (7) is connected to VDD (8); SCL (10) is connected to SCL (6) of the PCA9536; SDA (9) is connected to SDA (7) of the PCA9536; VCC (8) of the PCA9536 is connected to VDD (8); and the output pins P0 (1), P1 (2), P2 (3), and P3 (5) are connected to PD2\_ext, PD1\_ext, PD0\_ext, and PD3\_ext respectively. A 0.1uF capacitor (C45) is connected between VDD (8) and GND. A 0.1uF capacitor (C46) is connected between VDD (8) and GND.

Rev:  
Id: 4/4